

JEE MAIN REVISION MASTER: DIFFERENTIAL EQUATIONS SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 244 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Differential Equations - Core Application	244 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Differential Equations - Core Application (Very High Yield — 244 PYQs)

Core Principle: Key problem pattern in Differential Equations.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x\sqrt{1-x^2}dy+(y\sqrt{1-x^2}-x\cos^{-1}x)dx=0, x \in (0,1), \text{underset}{x \text{ right...}}$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $(dy/dx)=(1+x+x^2)(1-y+y^2), y(0)=(1/2)$. Then $(2y(1)-1)$ is equa..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x(dy/dx)-y=x^2\cot x, x \in (0,\pi)$. If $y(\frac{\pi}{2})=\frac{\pi}{2}$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let $y=y(x)$ be the solution of the differential equation $(dy/dx)=(1+x+x^2)(1-y+y^2)$, $y(0)=(1/2)$. Then $(2y(1)-1)$ is equal to

- (A) $\sqrt[3]{3}\tan(\frac{1}{\sqrt[3]{3}})$
- (B) $\frac{\sqrt[3]{3}}{2}\tan(\frac{1}{\sqrt[3]{3}})$
- (C) $\sqrt[3]{3}\tan(\frac{1}{\sqrt[3]{3}})$ **[Correct]**
- (D) $\frac{\sqrt[3]{3}}{2}\tan(\frac{1}{\sqrt[3]{3}})$

Correct Answer: C

Solution Analysis: Given $(dy/dx)=(1+x+x^2)(1-y+y^2)$, $y(0)=(1/2)$ We will solve it by the method of separation of variables. Since $\frac{dy}{1-y+y^2}=(1+x+x^2)dx$ Now simplify the denominator on the left: $1-y+y^2=y^2-y+1=(y-\frac{1}{2})^2+(\frac{3}{4})$ So, $\int \frac{dy}{(y-\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}=\int (1+x+x^2)dx$ Using the standard result $\int \frac{du}{u^2+a^2}=\frac{1}{a}\tan^{-1}(\frac{u}{a})$ we get $\frac{2}{\sqrt{3}}\tan^{-1}(\frac{2y-1}{\sqrt{3}})=x+\frac{x^2}{2}+\frac{x^3}{3}+C$ Now use the initial condition $y(0)=(1/2)$. Then $2y-1=0$ so